

Alan Turing “Computing Machinery and Intelligence”, *Mind*, 1950. p. 460

It is a pleasure to me to direct your attention on the recent work of this brilliant author¹, that is less influential than he should be. He is a mathematician, but in this work the topic is not mathematics or formal tools, but a much more challenging issue: a principled foundation of "cognition" and of its "machinery" and function, and thus the foundations of CS and of AI; back to the future.

I wish that this paper (and Turing's previous works) have the impact they deserve, in order to bring back especially AI to reflect on its nature and mission. Now the dominant feeling and view in AI is that what they really are is "engineers" and what they have to do is produce "technics", software solutions of practical problems with a relevant industrial, commercial, or military impact. AI was and (involuntarily) is a science, aimed at "understanding", modeling, building (computational) theories of (possible) cognition, interaction, society,² I understand that research (not researchers) needs money, but research should never forget its *basic questions*.

Turing - without any aggressiveness - tries to bring us back to our role and origin, and to reflect on our (AI & CSs) theoretical and methodological foundations.

His raising the AI challenges and reflection on that is very timely, since AI (not "abortive" at all, as a lot of people - while taking/exploiting the AI solutions - like to claim) is quite threatening in its results and perspectives. And the major threat is precisely the lack of awareness and critical discussion.

AI cannot continue - only for condescension or convenience - to disguise itself just as a "technology" or to deny the radical character of its challenges. AI should have more critical awareness and be more explicit on what it does or can do (or not); not just negation or removal.

I also wish that the author will not be irritated, but I do not pretend (nor intend) to give the correct and - even less - the complete interpretation of his main results and ideas. I just want to present and discuss *some* issues I was very impressed by and possible exciting digressions.³

1. Must "intelligence" just be "natural"? The AI R-Evolution

It is true that “A great positive reason for believing in the possibility of making thinking machinery is the fact that it is possible to make machinery to imitate any small *part* of a man”, that is, also perception ("sense organs"), motor action and environment changing, sex, .. We have in fact "robots" with that and guided by artificial intelligences; but the problem is a bit different and more radical. Will we have "intelligence" only if/when we will root cognition in that perceptual and embodied substrate? The author should more explicitly and directly discuss the "embodiment" very trendy approach which is criticizing the "computational" view of cognition, and even the "functionalist" view. Can "mind" be there without a "body"? (Not of course without a material implementation). What kind of mind or what aspects of mind?

Questions are in fact more radical. Isn't "intelligence" just an adaptive expression of "life"? That is: Is life a *pre-condition* for any form of real intelligence? Can we have intelligence just as reasoning, without practical action in a word (computers) and the needed body? And can we have a real body and real behavior (robots) without life?

¹ I'm sorry with the journal because I'm late with my contribution; but I was waiting for text of the Radio interview of the author, that was coming.

² Escaping not only from a scientific role, but from more demanding commitments and philosophical, political, moral, and scientific issues (see below).

³ I never had the opportunity of meeting Turing personally, and to discuss directly with him; but I'm sure that this will happen soon; I wish so because there are some issues that I would discuss vis a vis with him, in my bad but more vivid spoken "English" than in my written one. So I take just for me some controversial issues; but I invite my reader to read Turing's paper because s/he will be really excited (or upset?).

Nature has proceeded in the opposite direction: life, purposive behavior, affects, choices, cognition and reasoning. We are doing - this is the crazy challenge of AI (Turing remind us) - a "r-evolution" in the sense of a "reversed evolution": we have started from superior cognitive functions, from abstract symbolic thinking, then relate it to goals and thus "motivation" for driving external effective "behaviors"; then we have introduced "learning" based on some association and reward; then (having the "motives") we are introducing "emotions" as reactions and as internal states; and then we can model a body, that means proprioception and enteroception, something we "feel", and (perhaps) even pleasure and pain; and finally we will (probably) build "living systems" with their reproduction mechanisms, niche, selection, autonomy and self-motivation, energy search and acquisition,⁴

Are the results of natural evolution "principled" or just "accidental" and accidentally adaptive to a given environment? Is - for example - just an evolutionary accident that our "motivation" is strongly grounded on emotions? Or we might - in principle - have really purposive, goal-directed, motivated actions in a "cold" mind and non-affective agent, able to plan, solve problems, choose, achieve, and learn from experience? I'm sorry but "Nature" is not providential, and one can in principle find different solutions. Science has to understand and model natural phenomena - of course - but also "normative models", possible worlds, underlying potential mechanisms and solutions.

Is AI a fantastic experimental, non-speculative but empirical and procedural, platform for designing and understanding the "possible" forms of action, intelligence, ... even separating what in nature is accidentally (but adaptively) intrinsically combined? Can be there *un-natural* intelligence and life?

Two years ago, in his essay *Intelligent Machinery* Turing considered "the question as to whether it is possible for machinery to show intelligent behavior" and recognised objections to the idea of intelligent machines "cannot be wholly ignored, because the idea of 'intelligence' is itself emotional rather than mathematical".⁵

In my view, a weak point of our author's vision is that there is not enough stress on the "goal-oriented" nature of mind, on its function to "control" in strict sense, to drive behavior towards ends; knowledge, reasoning, decision, ... are just for that. The relation between intelligence and goals is primary also respect to the relation with affects, emotions, ..

Notice (this is relevant for a discussion that persecutes us since 50 years) that the author also gives us a more realistic view of cognitive systems and functions; not as a "Yes/No" "All or Nothing" ("It is Intelligent or is not"; "Computer already think!", "Computer will never think; It is impossible in principle!"). Aren't there different components and functions and thus *forms* and *degrees* of intelligence and thinking? Shouldn't be there partial and progressive positive answers? In this sense Turing remarks the possibility to add the "parts of a man" (senses, action, speaking, but also "cognitive" skills); piece by piece but in an appropriate architecture of layers and components.

I also like that he refuses a generic discussion about "are/can be machines intelligent" based on a discussion of how "intelligence" should be defined. This is a tricky discussion, since several people give on purpose more and more restricted definitions of "intelligence" just shaped for humans and excluding any other possible (also natural) forms, in order to conclude that machines cannot - in principle - be intelligent. A mere tautology!

2. "M or F?"; intelligence is deception

⁴ There is also since several years an attempt to partially correct this path, to have a more "bottom-up", sub-symbolic approach (Brooks, ALife, neural nets, ...) building intelligent behaviors without "reasoning" (symbol manipulation); however, also in these cases one starts from behavior and interaction with the environment and learning, but not from synthetic "living" systems; at best can "simulate" them as living.

⁵ In Copeland, 2004: p. 411.

A special attention deserves a hint of this author, that I'm sure will be taken (too) seriously while it is just a simplification and a provocation (a game)⁶, which however has some deep meanings. Let's take some fun.

Is there a specific affinity between AI and deception? Is deception ability the real "test" of an effective AI and perhaps its real mission? Is (reciprocal self-)deception the real test of sociality?

Our author has fun by ridiculing our incapacity in a H-C interaction to recognize a real human, to distinguish whether we are interacting with a person or a "machine" (an artificial intelligence/agent). He proposes a "game" (imitation game), which is intrinsically social, interactive and aimed at deceiving.⁷

As I said, if I was Turing I would be more careful in making clear that it is an anti-skeptical divertissement, a teasing; I worry that it will be taken very seriously by philosophers and several scholars, with useless disputes and demonstrations. Actually this "game" is every day played on the WEB by billions of people and the so called "test" is passed (as *implicit*) in millions of interactions and messages exchange with artificial systems; what is amazing is precisely that for a lot of users this is *not relevant*: we do not care to understand whether the answer is from a human operator or from a machine. In fact there is not such a big difference, also considering that both of them can try to deceive us, and we must trust both for interacting.

There is another risk in the proposed example: that it is centered on linguistic interaction, as if this was the sign or essence of intelligence and of sociality. For sure several people will distort its interpretation in this direction.

It should be clear that the same game might be played on *practical* cooperative or competitive behaviors, on social "games" (like scripts and roles), or in emotional interaction based on expressions or reactions.

The real meaning of the "game" - to me - is primarily "methodological" and it is very clear: *What the real meaning of that question ("Can machines think?"): let us answer to that question as we implicitly answer to it in everyday life and interactions!* "Likewise according to this view the only way to know that a man thinks is to be that particular man. It is in fact the solipsist point of view. ... A is liable to believe 'A thinks but B does not [NdR. How in fact to prove that B thinks?]' whilst B believes 'B thinks but A does not'. Instead of arguing continually over this point [NdR. or ask for demonstrations] *it is usual to have the polite convention that everyone thinks*".

Let's be polite also with our social machines! We are in fact just building/evolving "a general educated opinion" and the appropriated broader "use of words" for *simply presupposing (like with people) that machines are thinking* in our living with them and through them.

The clear claim is that: like we *ascribe* mental contents and processes to humans on the mere basis of their performances and interactive responses, the same we (willing or unwilling) necessarily do with computers, and are unable to distinguish them from human partners.⁸

Intentional stance

⁶ People already started to propose this "game" as the "test" of machine intelligence, and call it "Turing test"! But this is not the sense of Turing's provocation, which is about how we in fact ascribe intelligence to people.

⁷ In the "game" there are three people, a man (A), a woman (B), and a interrogator (C) – either a man or a woman; C is closed in a room and communicates with A and B by a telewriter, and his/her goal is to determine which of the two is male and which is female. The goal of A in the game is to deceive C and induce him/her to a wrong identification; while the goal of B is to help C find the truth. Both A and B have to answer C's questions. As Turing notices, B might also use such expressions as "Don't believe him! I am the woman", but this will not help very much since A too could say something like this.

At this point Turing raises the following question: What will happen when/if a computer will take the place of A, of the deceiver? Would C be wrong as frequently as in the previous situation? *And he claims that those questions are equivalent to the question "Can machines think?"*.

⁸ *Il vero problema è che sarebbe ora di smettere di "fare finta" e di confondere le idee alla gente con questo. Come accade nell'ambito dei sistemi emotivi. Come ci sono sistemi che ragionano davvero e risolvono problemi, così ci possono essere e si cominciano a fare sistemi con una reale dinamica emotiva interna, che regola la loro condotta; non semplici maschere, che tuttavia già ingannano perfettamente!*

In which sense for example shouldn't we "*recognize*" a mind to an interactive entity able to help us by understanding what we have in mind ("mind reading"), what we are doing or would like to do, what we need without an explicit request? This is the same reason and way we ascribe a mind to "people" ("intentional stance").

As we said intelligence is socially "ascribed" as a fundamental presupposition for "socially" (not physically) interacting.

"The extent to which we regard something as behaving in an intelligent manner is determined as much by our own state of mind and training as by the properties of the object under consideration ... with the same object therefore it is possible that one man would consider it as intelligent and another would not; the second man would have found out the rules of its behaviour". (p. 431).

Turing should also cite the very nice results of XXXXX experiment, where the child while playing with and speaking to the robot dog is absolutely sure that it is understanding him and interprets the behavior of the dog as responses to his words and as playing with him. This is our spontaneous attitude. Don't forget that even scholars were deceived by Eliza (the therapist machine) and were sure that it was "understanding" what the patient was saying and reacting in an empathetic way.

Where artificial intelligence really "lies"

Let's put also in another way: AIs, computational systems, computer mediated interactions, interfaces, robots, .. deceive us; even not on purpose or intentionally. They deceive us "functionally", as a way of working given our mental frame and search for affordances. We are mistaken, deceived, and we couldn't not (and perhaps want) deceive us; since we need to adopt an "intentional attitude" and ascribe mental states. This creates a (vicious? virtuous?) circle, which reinforces this spontaneous and unconscious tendency to self-deception (the baby-dog example, the Eliza case, etc.).

The designer and the producer of the artificial interactive systems just exploit (do not create) that.

Let me conclude this sub-section by insinuating that deception is a nice vocation for AI; it is its origin and deep challenge. We for example mentioned believable agents, but, after all, isn't "believability" just a form of deception? the more believable the more deceptive? But I also like (although Turing should discuss a bit the false-emotions expression in Agents and the "believability" literature) that Turing centers the deceptive strategy on communication and linguistic interaction, question-answering, and especially on the *attribution* of "intelligence" to the other.

In a sense we could say that deception is the original, foundational challenge and *proof* for AI. Will artificial systems enter into our normal/deceptive social conventions for interaction?

The impossibility litany

It is important that Turing be so clear because after 60 years from the foundation of AI and ALIFE there is still this boring discussion about the impossibility⁹ of an artificial brain, or a "child machine" (Turing), of a computational intelligence; a debate blind to what is actually realized or in project. Machine playing chess at the top level, demonstrating theorems, making good medical diagnoses, understanding language, interacting appropriately, appropriately expressing emotions and reacting to them, be pro-active and understanding what we need, negotiating and arguing, etc. etc. Clearly Turing has in mind "Deep Blue" able to win against the chess word champion, with very smart strategies; and "Watson" the IBM system (answering to natural language questions) that recently in an historic exhibition match on the television quiz show Jeopardy!, was prevailing against the best Jeopardy! players of all time.¹⁰ Against these repetitive litany about impossibility, or failure, ... Turing proposes that *deception* game, not to be taken seriously (or better to be taken more seriously, like any game and deception).

⁹ See on that hot discussion Yurij Castelfranchi's (2004) interview to Turing on AI history.

¹⁰ However, he is not yet satisfied; he - correctly - wants also a more "practical" and embodied intelligence, acting in the world (to see, to react, to act, to learn from direct experience), and a more complete social intelligence (mind ascription, initiative, ..).

Our question is: Isn't claiming such "impossibility" just a matter of jinx and good-luck, since they worry about an immoral or politically dangerous AI and its objective as un-ethical? It wouldn't be better to explicitly discuss about the ethical aspects and political problems than just repeating "Anyway it is impossible!".

Social engineers

Let us put in other words the real message of Turing, which is very basic, direct, and very current:

- *we (engineers) are building not just new technical systems, but new socio-cognitive-technical systems: new forms of intelligence and of sociality.*

Right. And - I would add - perhaps a better involvement of social and cognitive scientists would be important, not just for importing and formalizing their theories (as ICT is trivially doing), but for better understanding what we are *unaware* building, to build it in a more conscious way; to change people (education, practices, interactions,...) and adapt them to the new culture; and also to develop new theories (by providing the social sciences new conceptual and experimental instruments; see § 4).

Also because "intelligence" is intrinsically "social" (this is the other message of the "game"): intelligence is social in its origins (evolutionary and developmental), in its function (intelligence is for power and dependence, for cooperation, exploitation, competition), in its instruments (language, argumentation,...); and in its "ascription" as presupposition of interaction. So an intelligent machine is a socially working (deceptively interacting) machine; either in competitively playing chess or in conversation.¹¹

Turing is right in stressing that we are not just building an "intelligent machine" (intelligent like humans? in a different way? It doesn't matter so much (except while using the AI "minds" as psychological models)). What we really - unexpectedly - have is a *"hybrid" new society*, where social interaction of any kind (information exchange and K institution building; commerce; confidence and friendship; organization; political movement¹²; sex ("Am I male or female?"¹³); ...) is not just between humans but "mediated" by artificial supporting systems, is "hybrid": that is, there are "social" interactions (in full sense), social networks, and collective behaviors *between Humans and AI Agents, H and Robots, between Ag-Ag, between R-R, between Ag-R, and even between H-H*. And - this is the message - it is practically impossible to be sure of the nature of your partners also because it doesn't matter; *it is not relevant*.

The system is radically hybrid, mixed, of "natural" intelligence (actually rather acculturated and artificialized) and "artificial" ones (actually rather acculturated and even affective). Is it clear that to be a "person" and a given kind of "partner" (seller or buyer, doctor, friend, priest, judge, ...) is in large part an "institutional" construction? Now we are building the new "institutional" characters and roles, and the new institutional "scripts" for interactions: the hybrid "games" of our future society.

Good message, mister Turing; and I like that you send it to us in such an ironic way.

M or F? Who am I?

¹¹ "We may hope that machines will eventually compete with men in all purely intellectual fields. But which are the best ones to start with? Even this is a difficult decision. Many people think that a very abstract activity, like the playing of chess, would be best. It can also be maintained that it is best to provide the machine with the best sense organs that money can buy, and then teach it to understand and speak English. . . . Again I do not know what the right answer is, *but I think both approaches should be tried.*" (1950)

¹² See for example the collaborative tasks (managed by mixed networks and MAS) analyzed in (Grosz 2013; and Grosz et al. 2012), like scheduling meetings, organizing travels, health care, rescue team support, ...

¹³ It is well known that, especially in chat lines and games, a lot of people change gender, and you are never sure whether your interlocutor is a woman or a man.

Dear prof. Turing I also have enjoyed that you do not just plays on the issue "Human or Machine?!" but you provoke with "Am I M or F?!" (an obsessive game on the WEB). I do politically like this, not just as a reflection on the real weakness of these seemingly strong gender "identities", but for reflecting on its strongly conventional nature, and on the arbitrary social inferiority of women and absurd persecution of gay people.¹⁴

The right answer is: "Neither a human nor a machine!" (a surpassed distinction!), or "Both!"; "Neither a man nor a woman!" or "Both a man and a woman".

In the current perspective we have, we look at what we are building and is emerging in a "black or white" traditional categorization: the right answer (real result) is a third one. We have to change our mind and perspective; being more creative and open-mind like professor Turing; changing our categories for understanding the socio-technical revolution that is shuffling us.

3. Not just Intelligence but also Autonomy in interaction

Autonomy is central to understand the "philosophy" of what is happening in computing and in HM interaction; not just abstract "intelligence", but intelligence for *initiative in interaction*.

Turing is right on that; in a sense we are beyond the limits of Lady Lovelace's (Ada Byron) famous truism, that the user may know at each instant what is being done. Our intention in building computers had been that of treating them like slaves, of giving them tasks that are completely specified; until now, he says, machines have been used only in this way; but must this remain the case forever? In fact, in distributed computing and especially within the Agents paradigm we are beyond Lady Lovelace's limits, and this comes precisely from their "autonomy": in evolving and learning, in directly accessing information from the world, in moving, planning and problem solving, in taking initiatives, in having their own goals, etc.

But pro-activity, initiative is a hot point for some guy. For example, the philosopher Karl Popper felt provoked while Turing was claiming (in a Radio interview) *"Tell me what - in your view- a computer cannot do ... and I will build one on purpose"*.

This upset a lot Popper: *"There is something in particular that computers have not: initiative! We cannot 'describe' initiative, although it is something that any small children and animal has."*

Strange claim - to me - (especially from a philosopher) that there is something that in principle *cannot* be clearly, operationally, defined and described. I do not think so at all. On the contrary I think not only that we can define/characterize "initiative" and in general "autonomy" in social interaction, but that we already have (this is what Turing is referring to). "Agents" that are "autonomous" self-regulated, developing, learning, reasoning, systems that have initiative (they are *pro-active*)¹⁵ in interaction with humans; for example, they give more or better information than required, or they interrupt us to inform us about some event relevant for what we are doing (Grosz, 2013). The artificial entities' *initiative* and autonomy (I'm sorry for Popper) is so growing and evident that we are warring about governing it; and we (in MAS systems) work on how to design "mixed initiative", "adjustable autonomy", and how to combine delegation and trust with "control" (Castelfranchi & Falcone, 2000).

Eventually, what we have are *two* proactive, autonomous, self-governing, learning, systems, interacting with each other, more and more similar. As we said, not only computers become different but competitive intelligences or human-like intelligences or even artificial "persons" with their rights; but humans on the other side become a distributed collective computer and will have distributed and *artificial* minds and bodies.

¹⁴ We should never forget that gay people were obliged to hid their nature, were condemned in courts and put in prison (like Oscar Wilde), were forced to "medical" care like taking hormones, and somebody even decided to commit suicide to stop this persecution.

¹⁵ Turing should cite on that at least Wooldridge and Jennings, 1995; see also Castelfranchi, C. 1995.

4. AI as Science and its interaction with sciences

I would claim - with a Turing inspired "*computational radical attitude*" - that there are no alternative at all to "computer modeling", to provide not just mathematical or formal models but computational ones, if we have to model the "proximate causes" of a given "phenomenon", and its superficial dynamics; the *underlying "mechanisms" that determine those behaviors*. We need even "synthetic" modeling, that is the material construction of the modeled entity to show how it actually produces the "predicted" behaviors/effects in interaction with the environment.

Turing is right: "The whole thinking process is still rather mysterious to us, but I believe the attempt to make a thinking machine *will help us greatly in finding out how we think ourselves*."¹⁶ Exactly. ¹⁷ *Computational/synthetic modeling* will be pervasive for modeling any hidden mechanism and "dynamics": from chemical reactions, to DNA, from evolution, to psychological mechanisms, to social, economic, historical simulations. This is the message and the *gift* that ICT and in particular AI (the first attempt in this direction for the CS) has to give to science.

And not just computational "models" and conceptual instruments for the theory, but also *experimental platforms*, new empirical data due to simulation, new predictions, etc. Including (but this is my claim) experiment that are impossible in "nature", either for practical, social, moral reasons (demography, urbanistic, etc.) or for the natural inseparability in human subjects between for example motivation and emotion. This will be crucial both for modeling "proximate causes" for the diachronic, evolutionary causes.

And also a fundamental device for intelligently *collecting and analyzing relevant data* (from the WEB, for example "big data", and from human behavior in natural conditions (traffic, investments, migration, ...)).

In sum, in a few years, no computing no science.¹⁸

References

Castelfranchi, C. Guaranties for Autonomy in Cognitive Agent Architecture. In M. Wooldridge and N. Jennings (Eds.) *Intelligent Agents*. Springer. LNAI 890, 1995, 56-70.

Castelfranchi, C. Falcone, R. Conficts within and for collaboration. In C. Tessier, L. Chaudron, HJ. Muller (Eds.) *Conflicting Agents: Conflict Management in Multi-Agent Systems*, Kluwer, Dordrecht, 2000, 33-61.

Castelfranchi, Y. Biting the apple: the challenge of Artificial Intelligence. Special Issue "A Mente Humana", *Multiciencia*, 3, Oct. 2004

Grosz, B. What Question Would Turing Pose Today? *AI Magazine*, winter, 2012, winter, 73-81.

¹⁶ Typescript, BBC Third Programme 15 May 1951; reprinted in Shieber, *The Turing Test: Verbal Behavior as the Hallmark of Intelligence*, 2004

¹⁷ However, a specific issue that Turing should deeply discuss is whether his "universal and recursive machine" (even including "a random element in a learning machine") is or not a good model for *brain* and the complexity of neural processing. This is a debatable issue.

¹⁸ Also in the sense that major scientific discoveries will be made by computer (able to manage Big Data, to demonstrate theorems, to interpret the laws and mechanisms of that data), but also in the sense of "human" traditional science supported by computational instruments.

Grosz, B. "A Turing MAS Challenge" Invited talk, AAMAS 2013, S. Paul Minnesota.